Bird Species Observation Dashboard Report

SQL + Streamlit Based Ecological Data Analysis

Introduction

This project presents an interactive and SQL-driven analysis of bird species observation data using modern data tools such as `Streamlit`, `SQLite`, and `Plotly`. The dataset contains detailed records of bird sightings across multiple locations, including species information, observation time, environmental conditions, and observer details.

The primary objective of this study is to analyze bird species distribution, behavior, and habitat preferences while understanding how environmental factors influence bird activity. The project also aims to identify biodiversity hotspots, seasonal trends, and conservation priorities using structured queries and interactive visualizations.

By integrating SQL analysis with a dynamic dashboard, this project enables efficient exploration of ecological data and supports data-driven decision-making for biodiversity conservation and environmental management.

Methodology

Data Source and Storage

The bird observation dataset is stored in an SQLite database named `bird_data.sqlite`. This database contains structured records of bird sightings collected across multiple administrative units and habitats. Using SQLite enables efficient querying and seamless integration with the Streamlit application.

Data Preprocessing

Before analysis, the dataset undergoes several preprocessing steps to ensure data quality and consistency:

- Date values are converted into datetime format to extract temporal features such as month and day.
- Temperature and humidity columns are converted into numeric format to enable environmental analysis.
- Missing or invalid values are handled using appropriate techniques such as coercion and filtering.
- Additional derived features (e.g., month, hour of observation) are created for deeper temporal analysis.

Tools and Technologies

The project utilizes the following tools and technologies:

- **Streamlit:** For building the interactive dashboard interface
- **SQLite:** For efficient data storage and SQL querying
- **Pandas:** For data cleaning and manipulation
- **Plotly:** For creating interactive visualizations

Analysis Approach

The overall workflow of the project follows a structured pipeline:

$$\text{Data Collection} \rightarrow \text{Data Cleaning} \rightarrow \text{SQL Analysis} \rightarrow \text{Visualization} \rightarrow \text{Insights}$$

This systematic approach ensures that the analysis is accurate, reproducible, and easy to interpret.

Dashboard Features

Interactive Filters

The dashboard provides a set of dynamic filters that allow users to customize their analysis based on specific criteria. These filters enhance usability and enable focused exploration of the dataset.

- **Administrative Unit Code:** Select specific regions or parks for localized analysis.
- **Month:** Filter observations based on time of year to study seasonal trends.
- **Species (Common Name):** Focus on specific bird species.
- **Location Type:** Compare habitats such as forest and grassland.
- **Temperature Range:** Analyze how environmental conditions affect bird activity.
- **Sex:** Study distribution based on gender (Male, Female, Unknown).

Overview Dashboard

The overview section presents key performance indicators (KPIs), including:

- Total number of observations
- Number of unique species
- Total number of observation locations (plots)

It also includes visualizations such as top observed species, providing a quick summary of biodiversity patterns.

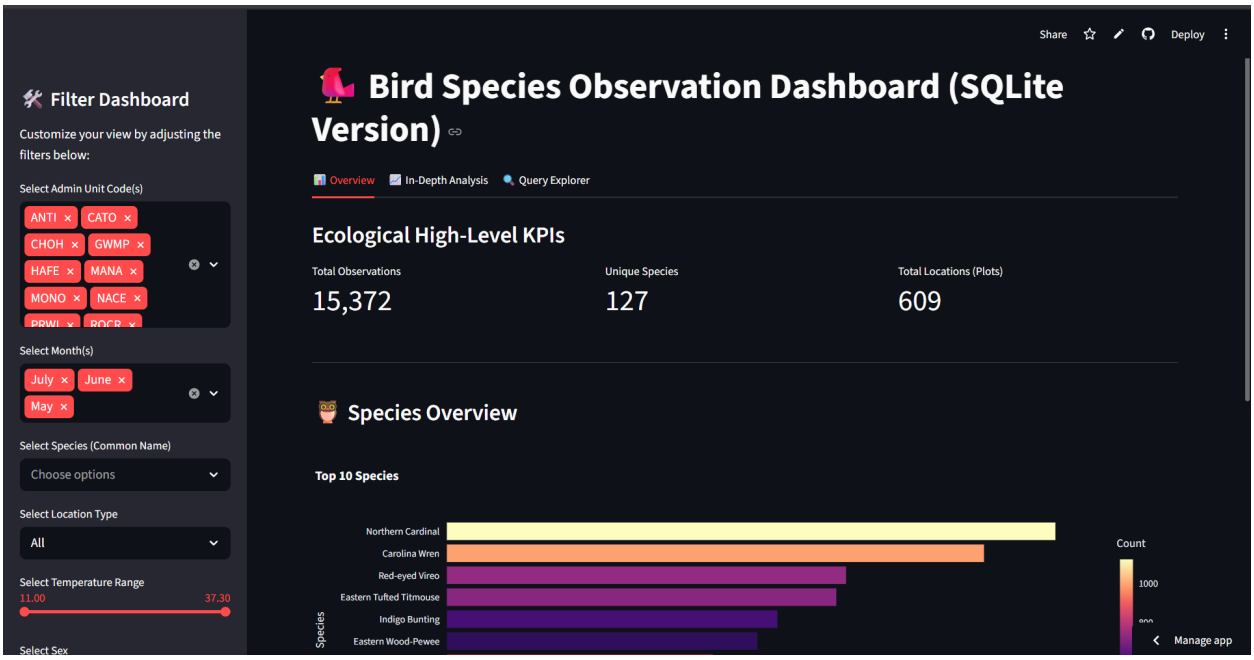

Figure 1: Bird Species Observation Dashboard Overview

In-Depth Analysis

The dashboard includes detailed analysis sections:

- **Temporal Analysis:** Observations by hour, month, and day
- **Spatial Analysis:** Habitat comparison and hotspot identification
- **Species Analysis:** Frequency and rarity distribution
- **Environmental Analysis:** Impact of weather conditions
- **Behavior Analysis:** Identification methods and bird activity
- **Observer Analysis:** Contribution patterns of observers

SQL Query Explorer

An advanced feature of the dashboard allows users to execute custom SQL queries directly on the dataset. This enables flexible data extraction, deeper analysis, and supports advanced users in exploring the dataset beyond predefined visualizations.

Key Analysis and Insights

Overview KPIs

The analysis begins with key summary metrics:

- Total number of bird observations
- Number of unique species recorded
- Total number of observation locations (plots)

Insight: These indicators reflect the overall biodiversity richness and spatial coverage of the dataset, forming the foundation for deeper ecological analysis.

Temporal Analysis

Observations by Time of Day: Bird activity peaks during early morning hours, commonly referred to as the *dawn chorus*, when birds are most active in feeding and vocalizing.

Observations by Month: Higher observation counts are recorded during spring and early summer, corresponding to migration and breeding seasons.

Observations by Day: Variation across days is influenced more by observer availability than biological patterns.

Insight: Temporal patterns highlight the importance of time-based factors in bird activity and observation efficiency.

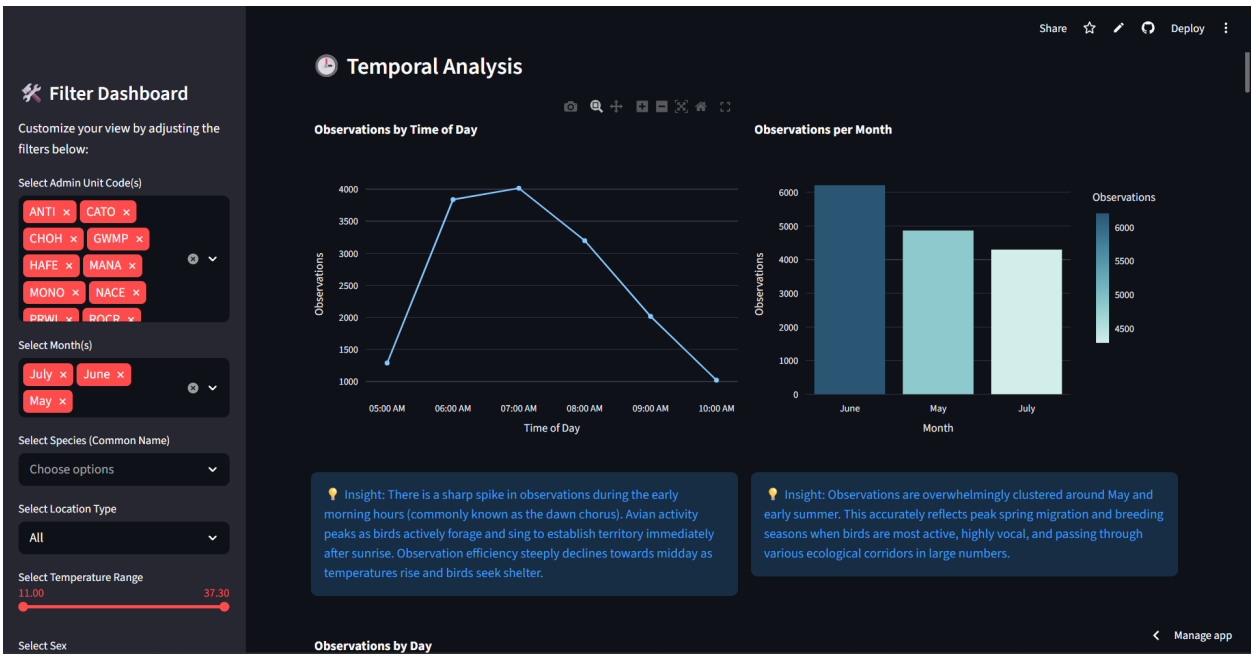

Figure 2: Bird Observations by Time

Spatial Analysis

Forest vs Grassland: Forest habitats show higher observation counts compared to grasslands, indicating either richer biodiversity or higher survey intensity.

Hotspot Analysis: A small number of plots contribute a large proportion of total observations, identifying biodiversity hotspots.

Insight: Spatial analysis helps identify critical habitats that require conservation attention.

Figure 3: Comparison of Bird Observations in Forest and Grassland Habitats

Species Analysis

Top Species: A limited number of species dominate the dataset in terms of observation frequency.

Rarity Distribution: Most species are observed only a few times, forming a long-tail distribution typical of ecological data.

Sex Distribution: Male birds are more frequently observed due to their visibility and vocal behavior.

Insight: The ecosystem consists of both dominant species and many rare species, highlighting biodiversity complexity.

Environmental Analysis

Temperature and Humidity: Bird activity is highest within moderate temperature ranges and higher humidity levels.

Sky Conditions: Clear and partly cloudy skies are associated with higher observation counts.

Wind Conditions: Calm weather conditions favor bird activity and detection.

Disturbance: Lower disturbance levels result in higher bird sightings.

Insight: Environmental factors significantly influence bird behavior and observation success.

Behavior Analysis

Identification Method: Auditory detection (bird calls and songs) is more common than visual identification, especially in dense habitats.

Flyover vs Ground Activity: Most birds are observed actively using habitats rather than flying overhead.

Insight: Behavioral patterns indicate strong habitat dependence and the importance of acoustic detection.

Figure 4: Impact of Environmental on Bird Observations

Observer Analysis

Top Observers: A small group of observers contributes a large portion of the data.

Insight: Observer bias may affect the dataset, and wider participation can improve data balance and coverage.

Conservation Analysis

PIF Watchlist Species: The dataset includes sightings of species that are considered at risk.

Regional Stewardship Status: Certain species require focused conservation efforts due to their ecological importance.

Insight: The analysis highlights critical species and habitats that should be prioritized for conservation planning.

Conclusion

The Bird Species Observation Dashboard successfully integrates SQL analysis with interactive visualization to provide meaningful insights into bird species distribution, behavior, and environmental interactions. By leveraging tools such as `Streamlit`, `SQLite`, and `Plotly`, the project enables efficient exploration of complex ecological data.

The analysis reveals clear temporal patterns, with bird activity peaking during early morning hours and specific seasons such as spring and early summer. Spatial analysis highlights the importance of forest habitats and identifies key biodiversity hotspots that contribute significantly to overall observations.

Environmental factors such as temperature, humidity, sky conditions, and wind play a crucial role in influencing bird activity and observation success. Behavioral analysis further emphasizes

the dominance of auditory identification methods and the importance of habitat-based activity over flyover observations.

Additionally, the study identifies observer contribution patterns and highlights potential biases in data collection. The conservation-focused analysis brings attention to at-risk species and regionally important species, supporting targeted conservation efforts.

Overall, this project demonstrates the effectiveness of combining data analytics, SQL querying, and visualization tools for ecological research. The insights generated can support biodiversity conservation, habitat management, and data-driven decision-making. Future enhancements may include geospatial mapping, predictive modeling, and integration of real-time data for more advanced ecological analysis.